SMART HOME AUTOMATION SYSTEM

A Project Report

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IE and IOT

Bachelor of Technology

in

Artificial Intelligence and Data Science

by

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CERTIFICATE

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This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We hereby declare that the project report **SMART HOME AUTOMATION**, submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the School of AI, Amrita Vishwa Vidyapeetham, Faridabad, India is a bonafide work done by us under supervision of Dr. Abhishek and Dr. Sakshi Ahuja .

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

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Abstract

In the pursuit of smarter, safer, and more energy-efficient living environments, this project presents a comprehensive Smart Home Automation System that integrates motion-based lighting, gas leak detection, and vibration monitoring. By leveraging sensors and micro-controller-based automation, the system aims to enhance household safety, energy efficiency, and situational awareness. The motion lighting module utilizes a Passive Infrared (PIR) sensor to detect the presence of individuals within its range. Upon detecting motion, the system automatically activates lighting, improving visibility in dimly lit areas and significantly reducing power wastage by switching off when no activity is sensed.

The gas detection component of the system is powered by an MQ2 gas sensor, which continuously samples ambient air to identify the presence of potentially hazardous gases such as liquefied petroleum gas (LPG). When the gas concentration exceeds a predefined safety threshold, the system promptly issues alerts through auditory or visual means, enabling swift preventive action. This feature is critical in mitigating the risks associated with gas leaks, including fire hazards and explosions, thereby ensuring the safety of home occupants.

To further strengthen the system's safety capabilities, a vibration sensor is employed to detect abnormal shaking or tremors within the home. This function serves a dual purpose—early warning of seismic activities and identification of unauthorized tampering with household objects or infrastructure. All vibration data are transmitted to the ThingSpeak platform, where users can view real-time trends and receive notifications for unusual activity. This multifaceted automation system exemplifies how IoT-based solutions can be effectively implemented to create secure, responsive, and intelligent home environments.

Acknowledgement

We express our deepest gratitude to everyone who supported us throughout the development of our project, **"Smart Home Automation System."** This project was a remarkable learning experience, and it would not have been possible without the guidance, encouragement, and assistance of many individuals.

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Chapter 1

INTRODUCTION

In today's rapidly advancing world, the demand for intelligent safety and automation systems within homes is becoming increasingly critical. Factors such as urbanization, hectic lifestyles, and the rise of nuclear families necessitate real-time monitoring and responsive mechanisms to ensure household safety. This project introduces a Smart Home Safety Monitoring System designed to enhance security, automation, and environmental awareness in residential spaces.

The system integrates three core functionalities: motion-activated lighting using a Passive Infrared (PIR) sensor [1], gas leakage detection via the MQ-2 gas sensor[2] , and vibration sensing using the SW-420 module [3]. These components are orchestrated through an ESP32 microcontroller, serving as the central processing and communication hub. Collectively, these sensors ensure a safe and responsive environment by detecting intrusions, hazardous gas leaks, and unusual vibrations. Notably, the vibration sensor's capability to identify tremors renders it potentially useful for early earthquake detection.

The collected sensor data is transmitted to the ThingSpeak IoT cloud platform via the ESP32's built-in Wi-Fi capabilities, enabling remote, real-time monitoring through any internet-enabled device. This feature allows users to receive alerts and take action even when they are away from home. The platform also supports data visualization, enabling users to track trends and analyze system behavior over time.

Recent studies have explored similar integrations. For instance, Desnanjaya et al.

developed an IoT-based home security monitoring system using embedded controllers, incorporating PIR sensors and gas detectors to monitor and send real-time notifications about home conditions[4]. Similarly, a study presented at the E3S Web of Conferences described a home security system utilizing PIR sensors and cloud-based alerts to enhance residential security [5]. Additionally, research published in the International Research Journal of Modernization in Engineering Technology and Science discussed a smart surveillance system that captures data upon detecting motion to improve home security[6].

This integrated solution not only advances home automation but also significantly improves safety—particularly for households with children, elderly individuals, or persons with disabilities who are more vulnerable to emergencies. With further development, this system can be expanded into smart city applications or integrated with AI for predictive safety analytics.

The Smart Home Safety Monitoring System presents an efficient, scalable, and cost-effective approach to enhancing residential safety and automation. By integrating motion detection, gas leak monitoring, and vibration sensing with real-time cloud connectivity via ESP32, the system offers timely alerts and actionable insights. Its adaptability makes it especially beneficial for vulnerable households and lays a strong foundation for future advancements in smart living and urban safety infrastructure.

1.1 Main Objective

1. **Motion-Activated Lighting:** Utilize a Passive Infrared (PIR) sensor interfaced with the ESP32 to detect human motion and automatically activate lighting for enhanced visibility and safety. The lights are automatically turned off when no motion is detected, promoting energy efficiency. This objective can also be used for security purposes like detecting the motion of an intruder.

2. **Gas Detection System:** Implement an MQ2 gas sensor connected to the ESP32 to monitor the presence of harmful gases such as LPG. When gas levels exceed a predefined threshold, the system triggers immediate alerts (e.g., buzzer or cloud

notification) to prompt quick response and prevent potential fire hazards.

3. **Vibration Detection:**Employ a vibration sensor (SW-420) integrated with the ESP32 to detect unusual tremors or movements. This helps in identifying seismic activity or possible tampering. The ESP32 transmits this data to the ThingSpeak IoT cloud platform for real-time monitoring, logging, and alert generation.

1.2 Motivation of the project

The project aims to enhance safety, energy efficiency, and real-time monitoring through smart automation. Motion-activated lighting reduces unnecessary power usage by detecting human presence. The gas detection system ensures early warning against harmful gases like LPG, preventing fire hazards. Vibration sensing helps detect tampering or seismic activity. Using the ESP32 and ThingSpeak platform, the system offers a cost-effective, IoT-enabled solution for smart homes and environments.

1.3 Problem Statement

Traditional safety and monitoring systems often lack features such as real-time alerts, automation, and remote monitoring, reducing their efficiency in detecting motion, hazardous gas leaks, and physical disturbances. Additionally, conventional lighting systems waste energy by remaining ON regardless of occupancy. There is a growing need for an affordable, smart IoT-based solution that not only enhances safety by monitoring motion, gas leaks, and vibrations in real-time but also improves energy efficiency by automating lighting based on motion detection. Such a system would enable remote tracking and immediate alerts, while optimizing energy use in homes, industries, and public spaces.

1.4 Literature Survey

With the increasing need for intelligent home automation and safety systems, numerous studies have explored the integration of sensors and IoT platforms for smart residential environments. The following literature provides insight into recent technological advancements and related works that have inspired the development of this project.

1.PIR-Based Motion Detection Systems

Desnanjaya et al. proposed an IoT-based home security monitoring system that integrates PIR motion sensors and embedded microcontrollers. Their system enabled real-time monitoring of home environments and provided instant notifications in response to motion detection events. This highlights the effectiveness of PIR sensors in enhancing residential security by detecting unauthorized entries and automating lighting to prevent accidents in low-visibility conditions.

2.Gas Leak Detection Using MQ-2 Sensor

Research in smart safety systems has shown the MQ-2 gas sensor to be highly effective in detecting combustible gases like LPG, methane, and smoke. In a study presented at the E3S Web of Conferences, a home safety system was developed using the MQ-2 sensor, interfaced with microcontrollers to monitor gas levels. The system triggered alerts and notifications when gas concentrations crossed a predefined threshold, thus preventing potential fire or health hazards.

3.Vibration and Seismic Monitoring

Vibration sensors, such as the SW-420, are utilized for detecting sudden movements or tremors that may indicate tampering or seismic activity. According to a study published in the International Research Journal of Modernization

in Engineering Technology and Science, smart surveillance systems using motion and vibration sensors can capture environmental disturbances and trigger responsive mechanisms. This capability is crucial for early warnings in case of earthquakes or suspicious activities.

4.IoT Integration and Remote Monitoring

The use of IoT platforms such as ThingSpeak significantly enhances the functionality of smart home systems. These platforms allow real-time data transmission, cloud storage, and visualization. Several studies have implemented ThingSpeak to track environmental parameters and notify users through cloud-based applications. These technologies provide scalability and user accessibility from any internet-enabled device, making home monitoring highly convenient and effective.

5.Comparative Studies and Gaps

While the above studies demonstrate isolated implementations of PIR sensors, gas detection, or vibration monitoring, fewer systems combine all three for comprehensive home safety. Additionally, most solutions focus on alert generation without advanced features like cloud data visualization or predictive analytics. The integration of all three safety mechanisms with cloud-based remote monitoring, as proposed in this project, addresses this gap and provides a holistic solution.

1.5 Existing Technology

Several technologies are currently available in the market to enhance home automation. These technologies focus on improving energy efficiency, convenience, and safety.

1. Smart Hubs and Voice Assistants

- **G**oogle Nest Hub with Google Assistant
- **A**pple HomePod with Siri
- **S**amsung SmartThings Hub
- Central control for all smart devices via voice or app

2. Smart Lighting Systems

- **R**emote control via smartphone
- **C**olor and brightness adjustment
- **S**cheduling and automation
- **M**otion-activated lighting

3. Smart Security Systems

- **S**mart video doorbells
- **M**otion detectors
- **I**ndoor/outdoor surveillance cameras
- **R**emote access and alert notifications

4. Smart Appliances

- **S**mart refrigerators, washing machines, ovens
- **A**uto diagnostics and maintenance reminders
- **R**emote monitoring and control
- **L**G ThinQ, Samsung SmartThings, Bosch Home Connect

5. Smart Entertainment Systems

- **V**oice-controlled streaming
- **S**mart display integration
- **M**ulti-room audio
- **A**mazon Fire TV, Google Chromecast, Apple TV, Sonos

Chapter 2

WORKING PRINCIPLE

2.1 Methodology



Figure 2.1: working principle flowchart

Motion Activated Lighting

The motion detection system leverages a PIR (Passive Infrared) sensor to identify movement within its detection range. These sensors work by sensing infrared radiation naturally emitted by warm-bodied entities such as humans or animals. When motion is detected, the PIR sensor outputs a HIGH signal through its digital output pin.

This signal is received by one of the ESP32's GPIO pin. Upon receiving the signal, the ESP32 registers the motion event and transmits the data wirelessly to the ThingSpeak platform using its integrated Wi-Fi module. The event is then visualized on ThingSpeak in the form of graphs or status updates, allowing for remote tracking and monitoring of motion activity.

Gas Detection System

The gas detection system employs an MQ2 gas sensor to identify the presence of combustible gases such as LPG, methane, and smoke in the environment. The MQ2 sensor detects variations in gas concentration levels and provides an analog output that corresponds to the detected gas level.

The analog signal from the MQ2 sensor is read by the ADC (Analog-to-Digital Converter) pin of the ESP32.When the gas concentration exceeds a predefined threshold, the ESP32 logs the event and transmits the data to the ThingSpeak platform via its built-in Wi-Fi module. On ThingSpeak, the gas levels are displayed in real-time graphs, allowing for continuous remote monitoring and timely alerts in case of hazardous conditions.

Vibration Detection

The vibration detection system utilizes an SW-420 vibration sensor to sense mechanical vibrations or shocks. The sensor provides a digital signal when it detects significant movement or vibration in its surroundings.

The digital output from the SW-420 sensor is connected to a GPIO pin of the ESP32. When a vibration event is detected, the ESP32 registers the occurrence and

sends the data to the ThingSpeak platform over Wi-Fi. ThingSpeak then visualizes the vibration events through graphs or updates, enabling real-time remote observation of abnormal physical disturbances.

Cloud Integration with ThingSpeak

All three modules are seamlessly integrated into a centralized data acquisition and monitoring system powered by the ESP32 microcontroller. The ESP32 handles both digital inputs (from the PIR and vibration sensors) and analog signals (from the MQ2 gas sensor), processing the sensor data in real time.

Using its built-in Wi-Fi capability, the ESP32 transmits this data to ThingSpeak, a cloud-based IoT analytics platform. ThingSpeak enables real-time visualization, time-series analysis, and event logging, which aids in proactive decision-making and incident response. This IoT-enabled setup not only delivers prompt alerts but also facilitates remote monitoring of environmental and safety conditions, making it ideal for deployment in residential, industrial, and public infrastructure settings.

2.1.1 Communication protocol Used

The ESP32 microcontroller uses the HTTP protocol to communicate with the ThingSpeak cloud platform, enabling real-time transmission of sensor data such as motion detection, gas levels, and vibrations. With its integrated Wi-Fi module, the ESP32 can send data to the internet without the need for additional hardware. It uses HTTP GET requests to upload sensor readings directly to ThingSpeak channels. HTTP is particularly suitable for this application because it is stateless, ensuring that each data transmission is handled independently; it is lightweight, ideal for small payloads like sensor values; and it is universally supported, ensuring compatibility with most web servers and IoT platforms. The ESP32 also receives HTTP response codes after each request, which provide immediate confirmation of whether the data transmission was successful. This makes HTTP a reliable and efficient method for connecting the ESP32 to cloud-based monitoring systems.

2.2 Transistor and op-amp application

2.2.1 Transistor as a Switch for MQ-2 Gas Sensor

In the gas detection system, a transistor in a common-emitter configuration is used as an electronic switch to control the operation of a buzzer. This configuration allows the transistor to function in two key states: cutoff and saturation. When no gas is detected, the ESP32's GPIO pin remains LOW, keeping the transistor in the cutoff region. In this state, no base current flows, the transistor remains non-conductive, and the buzzer stays OFF. When the gas sensor detects a harmful gas, the ESP32 outputs a HIGH signal to the transistor's base via a current-limiting resistor. This forward-biases the base-emitter junction, driving the transistor into saturation. As a result, the transistor conducts fully, current flows from collector to emitter, and the buzzer is activated. This method allows the ESP32 to switch high-current loads efficiently without directly driving them, thereby protecting the microcontroller while maintaining reliable operation of the alert system.

2.2.2 Transistor as a switch for PIR motion sensor

In the motion-activated light circuit, a BC547 NPN transistor is used in a common-emitter configuration and functions as an electronic switch to control the LED. The transistor operates in two key states: cutoff and saturation. When no motion is detected, the input signal to the base of the transistor remains LOW, keeping the transistor in the cutoff region. In this state, no base current flows, the transistor remains non-conductive, and the LED stays OFF.

When motion is detected, a HIGH signal is applied to the transistor's base through a current-limiting resistor. This forward-biases the base-emitter junction, driving the transistor into the saturation region. In saturation, the transistor conducts fully, allowing current to flow from the collector to the emitter. This current passes through the LED, turning it ON. This switching mechanism enables efficient control of the LED, handling higher current loads reliably without continuous manual intervention.

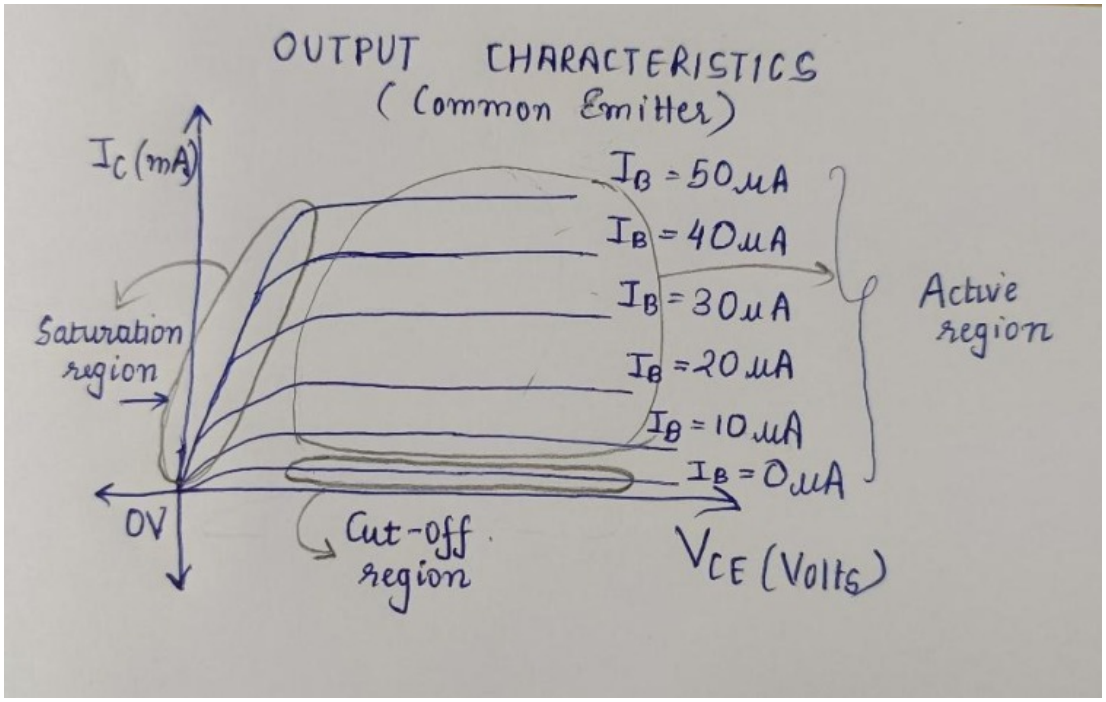


Figure 2.2: output characteristic graph

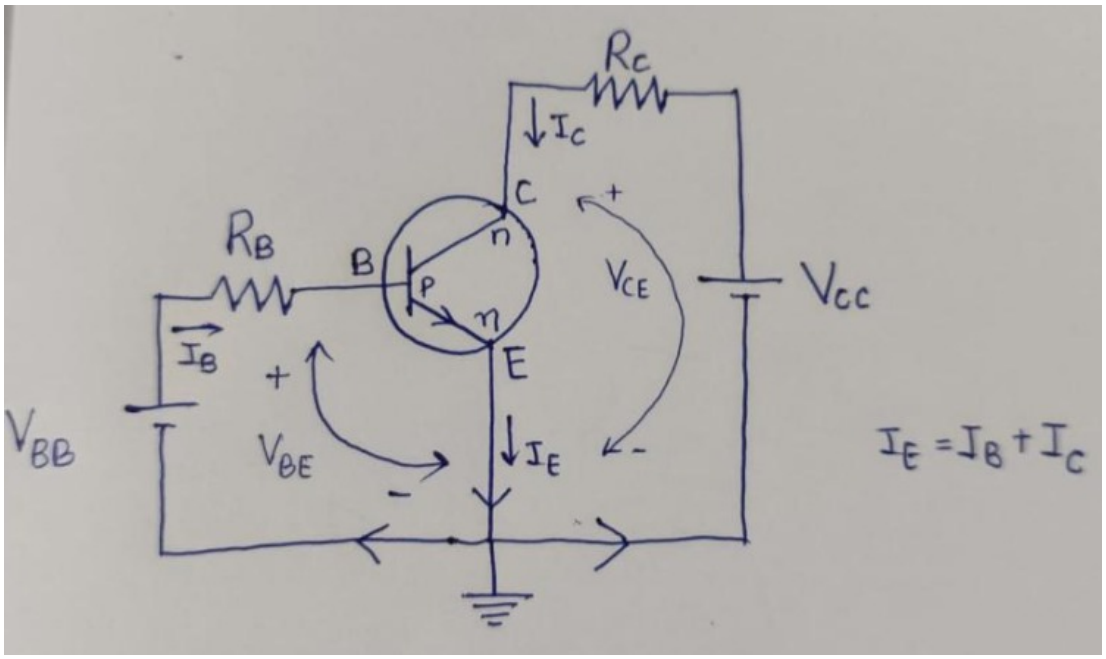


Figure 2.3: common emmiter configuration

2.2.3 Transistor and op-amp used in PIR Sensor

In the HC-SR501 PIR motion sensor, the internal circuitry includes an amplifier, a comparator, and a transistor, each playing a crucial role in accurately detecting motion

and providing a usable output signal.

- **op-amp as Amplifier:**

  The PIR sensing element produces a very weak analog voltage signal when it detects changes in infrared radiation. This signal is too small to be directly processed, so an amplifier (usually an op-amp) is used to boost the signal strength without distorting it. This ensures even slight motion can be detected effectively.

- **op-amp as Comparator:**

  After amplification, the signal is passed to a comparator, which compares the amplified signal to a preset reference voltage (often adjustable via the onboard potentiometers). If the signal exceeds this threshold—indicating motion—the comparator switches its output from LOW to HIGH. This digital decision-making step ensures that only significant motion triggers the sensor, reducing false alarms.

- **Transistor:**

  The final stage of the sensor includes a transistor that acts as a switch. It receives the digital output from the comparator and uses it to control the voltage at the OUT pin. When motion is detected, the transistor is driven into saturation, allowing current to flow and pulling the OUT pin HIGH. When no motion is detected, it enters the cutoff region, keeping the OUT pin LOW. This switching mechanism enables the HC-SR501 to interface cleanly with microcontrollers and external circuits.

Together, these components work in sequence to amplify, evaluate, and output a reliable digital signal in response to motion, making the HC-SR501 highly effective for automation applications.

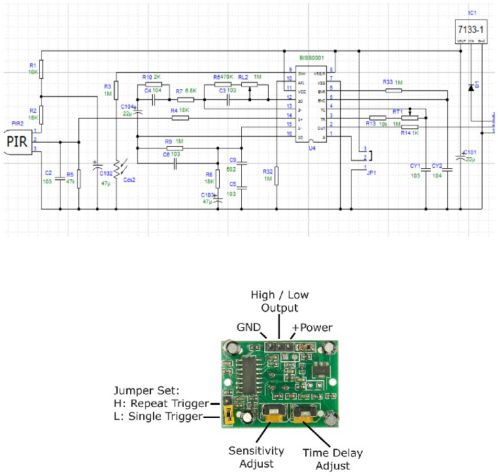


Figure 2.4: PIR Sensor Configuration

2.3 Transistor and op-amp used in sw-420

2.3.1 Role of Op-Amp (Comparator)

In this circuit, the LM393 is used as a comparator, which is a special type of operational amplifier (op-amp) designed to compare two voltages. It takes input from the SW-420 vibration sensor on one terminal and from a reference voltage set using a potentiometer on the other terminal. When the voltage from the sensor exceeds the reference voltage (indicating vibration), the LM393 output goes LOW, which acts as a digital signal to trigger the next stage of the circuit.

2.3.2 Role of Transistor

The 2N7002PW used here is an N-channel MOSFET transistor that acts as a switch. It is connected to the output of the LM393 comparator. When the comparator detects vibration and its output goes LOW, it turns ON the MOSFET, allowing current to flow through connected components like an LED or relay. This transistor effectively amplifies the small logic-level signal from the comparator and enables control of higher current devices, making it crucial for interfacing with external loads.

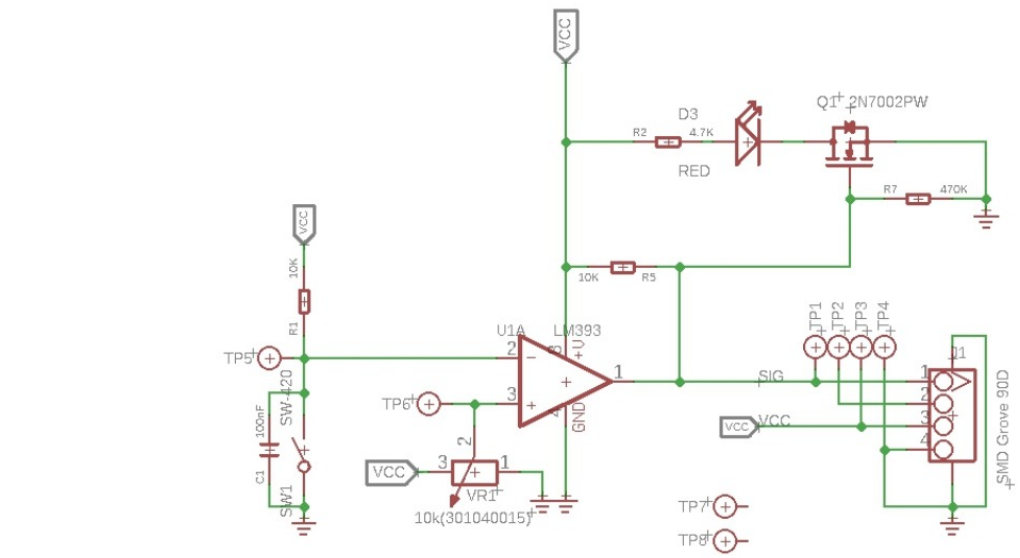


Figure 2.5: Transistor and op-amp inside sw-420

2.4 Hardware

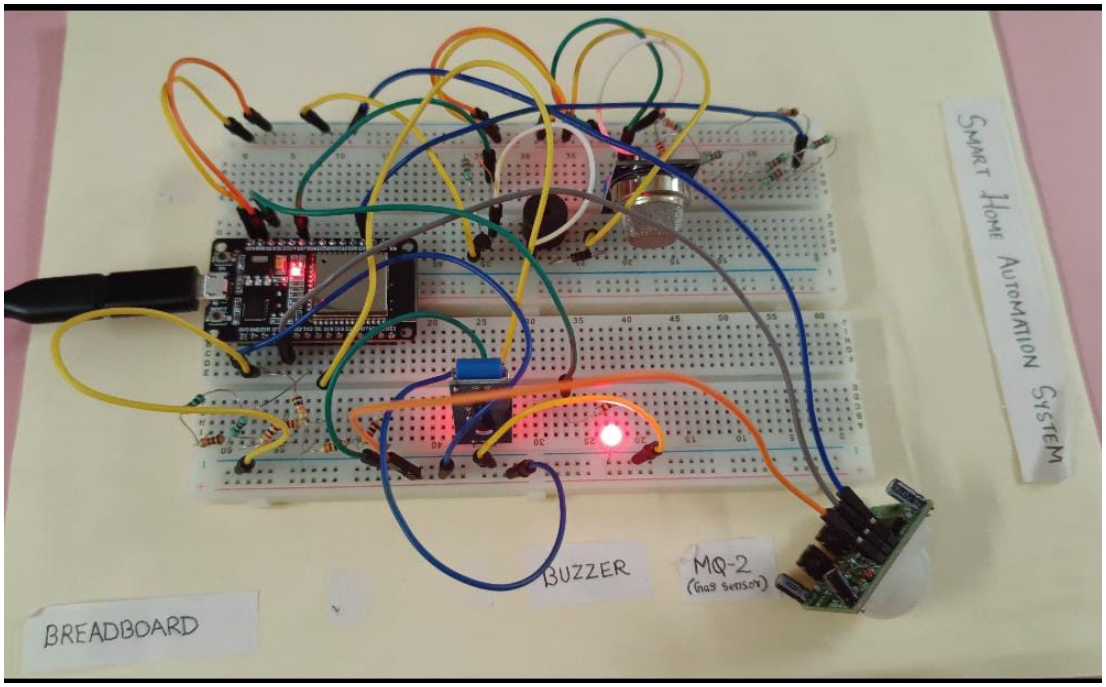


Figure 2.6: Smart Home Automation System Circuit

2.5 Simulation

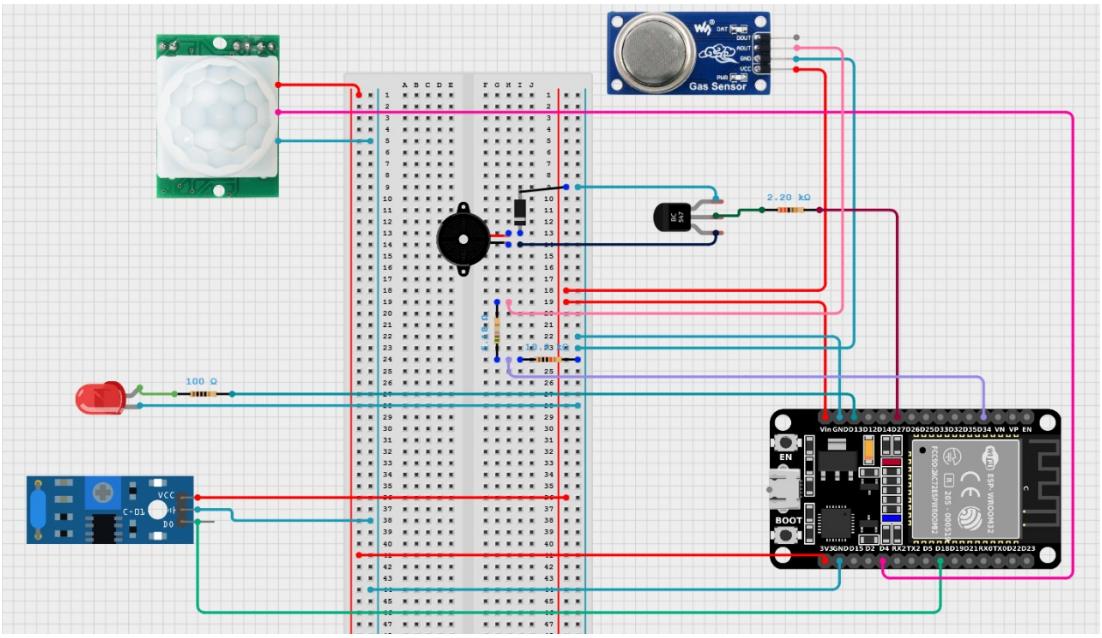


Figure 2.7: Circuit Simulation of Smart Home Automation System

2.6 Selection of Senors

2.6.1 MQ2 Gas Sensor:

We used the MQ2 gas sensor in our project because it is capable of detecting a wide range of combustible gases such as methane, LPG (liquefied petroleum gas), smoke, and hydrogen. This makes it ideal for preventing gas leakage hazards in residential environments, ensuring safety through early detection. Although other gas sensors like MQ5, MQ135, and MQ9 are available, we chose MQ2 because it offers high sensitivity to methane and LPG specifically, which are the most common gases used in homes. Its analog output is easy to read with the ESP32's ADC pins, and it is cost-effective, making it suitable for our smart safety system.

2.6.2 PIR (Passive Infrared) Sensor:

We selected the PIR sensor because it is excellent at detecting the presence and motion of humans or warm-bodied objects by sensing changes in infrared radiation. This was essential for our motion detection module, where we wanted to detect human

movement to trigger actions like lighting an LED and logging activity to the cloud. Alternatives like ultrasonic sensors (HC-SR04) or microwave motion sensors exist, but PIR was preferred because it is energy-efficient, requires no direct contact or reflection, and is less prone to false triggers indoors. Its simple digital output also makes it easy to interface with the ESP32.

2.6.3 SW-420 Vibration Sensor:

The SW-420 vibration sensor was chosen to detect mechanical vibrations or shocks that might indicate tampering, intrusions, or other physical disturbances. Initially, we considered using analog accelerometers like ADXL335 or MPU6050, but interfacing them with the Raspberry Pi posed challenges due to the need for analog-to-digital conversion and additional libraries. To keep the design simple and cost-effective, we opted for the SW-420, which is a digital sensor and easily connects to the ESP32 GPIO. Its reliability in detecting sharp movements, along with its low cost and ease of use, made it a practical choice for our vibration monitoring module.

Component	Operating Voltage	Current Requirement	Use / Purpose
MQ2 Gas Sensor	5V DC	~150 mA (preheat), ~50 mA (normal)	Detects **LPG, methane, smoke** to prevent gas leakage.
HC-SR501 PIR Sensor	5V DC	~50 µA (idle), up to 100 µA (active)	Detects **human motion** via infrared radiation.
SW-420 Vibration Sensor	3.3V – 5V DC	~10 – 15 mA	Detects **vibrations/shocks** for tamper or disturbance detection.
Active Buzzer	3.3V – 5V DC	~20 – 30 mA	Produces **sound alert** on gas leakage or vibration detection.
Red LED	2V (with 330Ω resistor)	~10 – 20 mA	Lights up to provide **visual alert** when motion is detected.

Figure 2.8: Component Specification

Table 2.1: Components Required

Hardware	**Cost (Approx.)**	**Specifications**
ESP32	500	The ESP32 is a low-power, dual-core 32-bit microcontroller running up to 240 MHz, with built-in Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 support. It has 520 KB SRAM, multiple GPIOs, ADC/DAC, and various interfaces like SPI, I2C, and UART. Operating at 3.0–3.6V, it features advanced power-saving modes and hardware security, making it ideal for IoT and embedded projects.
PIR Sensor	100	Detects infrared radiation emitted by the human body. Triggers output when motion is detected and is used in motion-activated systems with low power consumption.
MQ-2 Gas Sensor	150	Detects gases like LPG, methane, CO, and smoke. Provides analog output proportional to gas concentration. Used in fire and gas detection systems.
SW-420 (Vibration Sensor)	249	Senses vibrations or physical shocks and provides digital output. Used in anti-tampering, earthquake detection, and security applications.
Transistors (HC547)	10	General-purpose NPN transistors for switching small electrical loads or amplifying signals. Common in control circuits.
Diode (IN4007)	5	A rectifier diode allowing one-way current flow. Can handle up to 1000V reverse voltage. Used in power and rectifier circuits.
LED	30–150	Light-emitting diode used for visual status alerts. Low voltage operation; available in various colors and brightness levels.
Resistors	60	Limits current flow in a circuit. Essential for voltage division and signal conditioning. Available in many resistance values. Resistors used in the project 5.1k ohm, 10 k ohm ,100 ohm, 220 ohm.
Jumper Wires	40	Flexible, reusable wires for connecting components on breadboard or microcontroller pins.
Breadboard	55	Solderless prototyping board for easy component connection using terminal strips.
Total Cost	**1399**	

Chapter 3

RESULTS AND DISCUSSIONS

3.1 Code

SmokeSENSORBlynkIoT.ino

```
#include <WiFi.h>
#include <HTTPClient.h>

// WiFi Credentials
const char* ssid = "realme 10 Pro+ 5G";
const char* password = "radhakrishna";

// ThingSpeak API Key
const char* apiKey = "FGIUEKI1NCB71B17";
const char* server = "http://api.thingspeak.com/update";

// Pin Config
#define MQ2_PIN 34
#define BUZZER_PIN 27
#define VIBRATION_PIN 18
#define PIR_PIN 4
#define LED_PIN 13

int gasThreshold = 275;

void setup() {
  Serial.begin(115200);
```

```
  pinMode(MQ2_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(VIBRATION_PIN, INPUT);
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(LED_PIN, LOW);

  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500); Serial.print(".");
  }
  Serial.println("\nConnected to WiFi!");
}

void loop() {
  // Gas Detection
  int gasValue = analogRead(MQ2_PIN);
  int gasStatus = 0;
  if (gasValue > gasThreshold) {
    digitalWrite(BUZZER_PIN, HIGH);
    gasStatus = 1;
```

```
48    } else {
49      digitalWrite(BUZZER_PIN, LOW);
50    }
51
52    // Vibration Detection
53    int vibration = digitalRead(VIBRATION_PIN);
54    int vibrationStatus = (vibration == HIGH) ? 1 : 0;
55
56    // Motion Detection with LED Control
57    int motion = digitalRead(PIR_PIN);
58    int motionStatus = 0;
59    if (motion == HIGH) {
60      digitalWrite(LED_PIN, HIGH);
61      motionStatus = 1;
62    } else {
63      digitalWrite(LED_PIN, LOW);
64      motionStatus = 0;
65    }
66
67    // Debug Output
68    Serial.println("Gas: " + String(gasValue) + " | Status: " + String(gasStatus));
69    Serial.println("Vibration: " + String(vibrationStatus));
70    Serial.println("Motion: " + String(motionStatus));
71
```

```
72      // Send data to ThingSpeak
73      if (WiFi.status() == WL_CONNECTED) {
74        HTTPClient http;
75        String url = String(server) + "?api_key=" + apiKey +
76                     "&field1=" + String(gasValue) +
77                     "&field2=" + String(gasStatus) +
78                     "&field3=" + String(vibrationStatus) +
79                     "&field4=" + String(motionStatus);
80
81        Serial.println("Sending to ThingSpeak: " + url);
82
83        http.begin(url);
84        int httpCode = http.GET();
85        if (httpCode > 0) {
86          Serial.println("ThingSpeak Response: " + String(httpCode));
87        } else {
88          Serial.println("Error in HTTP request");
89        }
90        http.end();
91      }
92
93      delay(15000);  // ThingSpeak update limit
94    }
95
```

3.2 Explanation of the code

1. Libraries and Configuration

- include ¡WiFi.h¿ – Includes the WiFi library to connect to the internet.
- include ¡HTTPClient.h¿ – Allows sending HTTP requests (used to send data to ThingSpeak).

2. WiFi Credentials

- **s**sid and password store your WiFi network name and password.
- **T**he ESP32 connects to this WiFi network in the setup() function.

3. ThingSpeak Configuration

- **a**piKey – Your unique ThingSpeak Write API Key to send data to your channel.
- **s**erver – ThingSpeak's update URL.

4. Pin Assignments

Table 3.1: ESP32 Pin Connections

Pin	Component
D34	MQ2 Gas Sensor (Analog Input)
D27	Buzzer (Output Alert for Gas)
D18	Vibration Sensor (Digital Input)
D4	PIR Motion Sensor (Digital Input)
D19	LED (Turns ON with Motion Detection)

5. setup() Function

- **I**nitializes serial communication at 115200 baud rate.
- **C**onfigures all pins as INPUT or OUTPUT.
- **C**onnects to WiFi and prints connection status.

6.1 loop() Function – Main Logic

- **R**eads analog value from MQ2 gas sensor
- **C**ompares it with gasThreshold = 275.
- **I**f gas value is higher:
- **T**urns on the buzzer.
- **S**ets gasStatus = 1.

6.2 Vibration Detection

- **R**eads digital signal from the vibration sensor.
- **I**f vibration is detected (HIGH), sets vibrationStatus = 1.

6.3 Motion Detection (PIR)

* **R**eads digital signal from the PIR sensor.
* **I**f motion is detected (HIGH), sets motionStatus = 1.
* **T**urns LED ON when motion is detected.

7. Sending Data to ThingSpeak

* **I**f WiFi is connected:
* **B**uilds a URL with all sensor values (field1 to field4).
* **S**ends the data using HTTP GET request.
* **D**isplays the response code (200 = success).

8. Delay

* **d**elay(15000); – Waits for 15 seconds before sending the next update.
* **T**his is required due to ThingSpeak's update limit (1 update every 15 seconds).

3.3 Results



Figure 3.1: Visualization from ThingSpeak Platform

As shown in Figure, **In the first graph**, its presenting the motion detection data from a PIR (Passive Infrared) sensor over time.

Explanation:

The X-axis shows the timestamps when the data was collected.

The Y-axis (motion) shows the motion status:

* **1.00** indicates motion detected.
* **0.00** indicates no motion.

Interpretation:

* The red line shows rapid transitions between 0 and 1, meaning that motion was detected and not detected at various intervals.
* This pattern indicates that the PIR sensor actively monitors an area and detects movement frequently during the time shown.

In the second graph, its representing the gas detection data from an MQ2 gas sensor over time.

Explanation:

The X-axis (Date) shows the timestamps when gas levels were recorded. The Y-axis (gas-value) shows the gas concentration in analogue values:

* Higher values (up to 1000) indicate gas presence.
* Zero or low values indicate no gas detected.

Interpretation:

* Sharp spikes indicate gas was detected at the beginning.
* The flat line shows no gas was detected afterwards.
* The environment became clean or stable after initial gas detection.

In the third graph,its representing the vibration data from the SW-420 vibration sensor over time.

Explanation:

The X-axis (vibration) shows the vibration level (0 to 1 scale).
The Y-axis (Date) shows the timestamps when data was collected.

* A value of 1.00 indicates vibration detected.
* A value of 0.00 indicates no vibration.

Interpretation:

* The red line stays at 1.00, meaning continuous vibration was detected during the displayed time.
* This indicates constant movement or disturbance in the environment being monitored.
* It suggests either ongoing vibration or a need to recalibrate the sensor for sensitivity.

RASPBERRY PI-5 OUTPUT

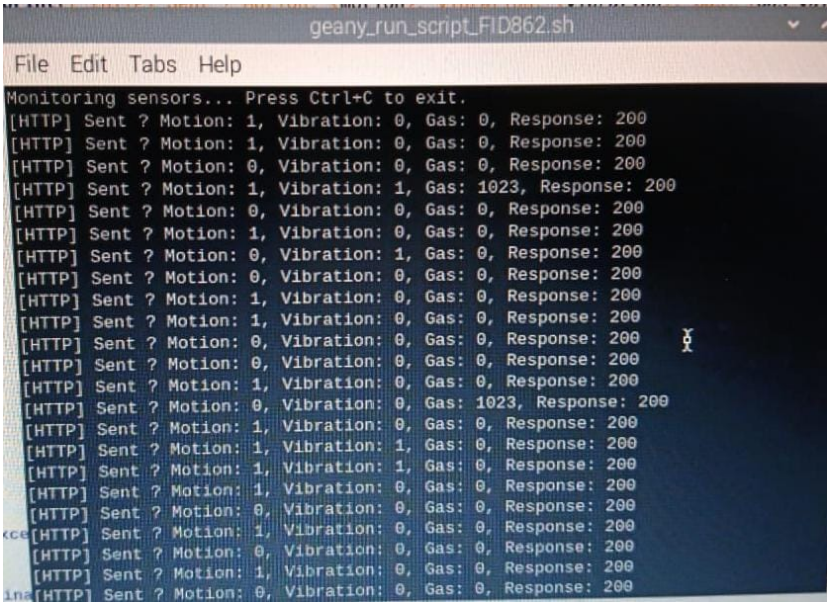


Figure 3.2: Real-Time Sensor Data Logging in Terminal

* **w**hen a lighter is brought close to the MQ2 gas sensor, the sensor detects the increased concentration of gas and immediately triggers the buzzer to turn on as an alert. At the same time, the MQ2 sensor measures the gas concentration level, which reaches a maximum value of 1023. This data is continuously transmitted to the ThingSpeak cloud platform, where it is recorded and visualized in real-time for monitoring purposes.

  When the vibration sensor detects any movement or vibration,The sensor's data is also sent to ThingSpeak, where it is logged and displayed for further analysis.

  When the PIR motion sensor detects motion within its sensing range, the system turns on the associated LED light to indicate detection. Simultaneously, the PIR sensor sends a signal that is recorded and uploaded to ThingSpeak, allowing for remote monitoring of motion activity.

3.4 Did we achieve our aim

Yes, we achieved our aim. We successfully built a Smart Home Safety System that includes motion-activated lights, gas leak detection, and vibration sensing, all connected through the ESP32. The system worked well and helped improve safety and automation in the home. It met our goal of creating a smart, affordable, and easy-to-use setup for better home protection.

3.5 Limitations

* **Limited Detection Range:** PIR and vibration sensors have a fixed detection range and may not effectively cover large areas or distant motion or vibrations.
* **False Positives/Negatives:**
  - PIR sensors can be triggered by pets or sudden temperature changes.
  - MQ2 sensors may respond to harmless gas concentrations, leading to false alarms.
* **Gas Sensor Calibration:** The MQ2 sensor requires periodic calibration to maintain accuracy. Poor or inconsistent calibration can result in unreliable gas readings.
* **Environmental Sensitivity:** Sensor accuracy can be adversely affected by environmental factors such as temperature, humidity, and dust.
* **Basic Alert Mechanism:** The current system relies solely on buzzers and ThingSpeak updates. It lacks mobile notifications, SMS alerts, or emergency response capabilities, which could improve safety and user experience.

3.6 Challenges Faced During the Project

3.6.1 MQTT Communication Challenges

During the initial stages, we opted to use the MQTT protocol to transmit sensor data because of its lightweight and efficient design for IoT applications. However, we consistently encountered error codes 4 and 5. Error code 4 typically indicates a connection refusal, while code 5 relates to authentication failure. These errors arose due to difficulties in establishing a stable and authorized connection with the MQTT broker, possibly caused by network configuration issues, broker restrictions, or incorrect credentials. Despite troubleshooting, the errors persisted and hindered reliable data transmission. To overcome this, we decided to switch to the HTTP protocol, which is simpler to implement and provided a more stable communication channel for sending data to the ThingSpeak cloud platform.

3.6.2 Transistor Functionality Adjustment

Initially, we planned to use the transistor as an amplifier to boost the sensor signals. However, since the transistor's output needed to be connected directly to the Raspberry Pi's GPIO pins rather than driving a buzzer or other high-power device, using it as an amplifier was not suitable. The amplified signals were not compatible with the Pi's input requirements, so we had to abandon the amplifier approach. Instead, we used the transistor as a switch, which provided a clear digital ON/OFF signal that the Raspberry Pi could reliably read and control, improving system stability and compatibility.

3.6.3 Voltage Level Compatibility Issue

Our sensors, particularly the gas sensor and vibration sensor, output signals at 5V levels. However, the Raspberry Pi's GPIO pins operate at a lower

voltage level of 3.3V and are not tolerant of 5V input signals. Feeding 5V directly into the Pi's pins could cause permanent damage. To address this, we incorporated a voltage divider circuit, which consists of two resistors arranged to step down the voltage from 5V to a safe 3.3V level before connecting to the Raspberry Pi. This hardware adjustment ensured safe and reliable operation without risking damage to the microcontroller.

3.7 Future Scope of the Project

Advanced Hardware Integration

- **Multi-Sensor Integration:** Incorporating additional sensors like temperature, humidity, or sound detectors will extend the system's capability to detect a wider range of environmental hazards such as fire or glass breaking.
- **Camera Module:** Integrating a camera (e.g., Raspberry Pi Camera or ESP32-CAM) enables capturing images or video upon motion detection. This enhances security by providing visual evidence.
- **Enhanced Output Devices:**
  - Smart relays can be used to control fans, alarms, or other appliances.
  - RGB LEDs can provide color-coded visual feedback for different alerts (e.g., red for gas leak, yellow for vibration, green for normal status).

AI-Based Detection and Analysis

1. **Pattern Recognition:** Machine learning algorithms can be employed to distinguish between normal and abnormal activity based on historical sensor data. This allows identification of patterns like repeated vibrations or gas spikes.

2. **False Alarm Reduction:** AI can help filter irrelevant events such as pet movement or minor gas fluctuations, ensuring that alerts are only triggered for legitimate threats.
3. **Predictive Maintenance:** Analysis of long-term sensor data enables prediction of hardware failures. This supports proactive maintenance, reducing downtime and improving reliability.
4. **Context-Aware Automation:** The system can respond differently based on context such as time of day, location, or specific user routines. For instance, lighting can adjust based on ambient light and occupancy.
5. **Computer Vision Integration:** AI-enabled image processing (facial recognition or object detection) allows for enhanced intrusion detection and monitoring.
6. **Adaptive Learning:** The system can improve its decision-making over time by learning from user behavior and feedback. This includes dynamic adjustment of alert thresholds and response strategies.

3.8 Comparison of Result and AIM

Our aim was to design and implement a smart home system that enhances safety and automation using motion detection, gas leak sensing, and vibration alerts, all controlled through the ESP32 microcontroller. We planned to make it affordable, user-friendly, and efficient for real-time monitoring.

The result closely matched our aim. We successfully developed a fully functional system where motion-activated lights respond accurately, the MQ-2 sensor detects gas leaks quickly, and the vibration sensor alerts in case of unusual movements. All components worked together smoothly, improving home safety and automation. The system also proved to be cost-effective and easy to use, fulfilling all our key goals.

Chapter 4

CONCLUSION

The project effectively integrates three key safety and automation features—motion lighting, gas detection, and vibration monitoring—into a compact, affordable, and intelligent home automation system. Each module is designed to respond to specific environmental triggers, thereby enhancing both the security and energy efficiency of residential spaces.

The motion lighting system, powered by a Passive Infrared (PIR) sensor, ensures that lights are only activated when human movement is detected. This significantly reduces unnecessary power consumption while improving visibility, especially during nighttime. The gas detection module utilizes an MQ2 sensor to continuously monitor for hazardous gases such as LPG. When gas concentrations exceed a predefined threshold, the system promptly activates an alert mechanism, preventing potential fire or health hazards. Furthermore, the vibration detection module identifies abnormal shocks or tremors, which could signal seismic activity or physical tampering. These events are logged to the ThingSpeak IoT platform, enabling real-time remote monitoring and historical trend analysis.

In unison, these components demonstrate how sensor-based automation can enhance modern living environments, making them smarter, safer, and more responsive. Potential future enhancements include deeper Internet of Things (IoT) integration, AI-driven analytics for pattern recognition and alert

optimization, and expanded applications in industrial, healthcare, and smart city contexts.

5.1 Summary

This project successfully integrates three vital functionalities—motion-activated lighting, gas leak detection, and vibration monitoring—into a unified home automation system. The motion lighting module uses a PIR sensor to detect occupancy and control lighting automatically, optimizing both convenience and energy usage. The gas detection system incorporates an MQ2 sensor to monitor harmful gases such as LPG, triggering audible alerts when dangerous concentrations are detected.

The vibration detection module adds a safety layer by sensing abnormal shocks or tremors and logging this data to the ThingSpeak cloud platform for visualization and analysis. Overall, the system offers an efficient solution for home safety and automation. Future developments could include the incorporation of IoT modules for mobile alerts, artificial intelligence for intelligent decision-making, and broader deployment in environments such as smart cities, healthcare facilities, and industrial settings.

Chapter 5

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